PHASE 1 – BRAINSTORMING & IDEATION INTRODUCTION

Project Title: EduGenie – Google Gemini Powered Learning Assistant

Brainstorming and ideation is the initial stage where the team identifies a real-world problem, discusses possible solutions, and develops a suitable project idea.

The team focused on common learning difficulties faced by students, such as difficulty understanding concepts, finding quick explanations, and getting interactive learning support.

Goal of this Phase: Identify an educational problem and generate possible technology-based solutions using AI.

Expected Outcome: A clear project concept that can be developed into a useful educational application.

Evidence Placeholder: Insert a screenshot/photo of the team's initial brainstorming discussion or project planning here.

PHASE 2 – PROBLEM IDENTIFICATION

Problem Identified: Students may not always receive immediate and personalized explanations when they have doubts while studying.

Key Issues Discussed:

• Difficulty in getting instant clarification of doubts.

• Searching through multiple learning resources can take time.

• Some explanations may be difficult for beginners to understand.

• Students may need examples or simplified explanations.

• Traditional resources may not provide an interactive question-and-answer experience.

Problem Statement: How can Generative AI be used to provide students with simple, interactive, and accessible learning assistance?

Evidence Placeholder: Insert a problem-identification notes/whiteboard screenshot here.

PHASE 3 – IDEA GENERATION

During brainstorming, the team generated several possible ideas for solving the identified learning problem.

Possible Ideas:

1. AI-powered educational chatbot.

2. Subject-wise question answering assistant.

3. AI quiz and practice-question generator.

4. AI-based study planner.

5. AI learning assistant with explanations and examples.

6. Voice-based educational assistant.

7. Personalized learning recommendation system.

The ideas were discussed based on usefulness, technical feasibility, ease of demonstration, and relevance to student learning.

Evidence Placeholder: Insert the team's idea list or brainstorming chart here.

PHASE 4 – DISCUSSION & IDEA REFINEMENT

The generated ideas were reviewed and refined by considering the needs of students and the available technology.

Discussion Points:

• The solution should be simple for students to use.

• The system should provide responses quickly.

• The project should demonstrate practical use of Generative AI.

• The solution should be possible to develop within the project timeline.

• The interface should be easy to understand.

After discussion, the team decided to focus on an AI-powered learning assistant that can answer educational questions and explain concepts.

Evidence Placeholder: Insert team discussion/idea refinement screenshot here.

PHASE 5 – FINAL IDEA SELECTION

Selected Project Idea: EduGenie – Google Gemini Powered Learning Assistant.

Why this concept was selected:

• It addresses a clear educational need.

• It provides interactive question-and-answer support.

• Google Gemini can be used for Generative AI-based responses.

• The idea can be demonstrated through a working application.

• The project can be extended with additional educational features in the future.

Core Concept: A student enters a learning question, the application processes the request through the configured Gemini service, and the generated explanation is displayed to the student.

Evidence Placeholder: Insert the selected-idea approval/planning screenshot here.

PHASE 6 – TARGET USERS & USE CASES

Target Users: Students and learners who need quick explanations or learning assistance.

Main Use Cases:

1. Student enters a question about a topic.

2. EduGenie processes the question.

3. Gemini generates an educational response.

4. EduGenie displays the response.

5. Student can ask another question.

Example Use Cases:

• Explain a concept in simple language.

• Give an example for a topic.

• Compare two concepts.

• Provide a short explanation or summary.

Evidence Placeholder: Insert a simple use-case diagram or planning screenshot here.

PHASE 7 – FEASIBILITY & PROJECT PLAN

Technical Feasibility: The project can be implemented using a programming environment, required libraries, a user interface, and Google Gemini/Generative AI integration.

Operational Feasibility: The proposed workflow is simple: enter question, submit request, and view response.

Development Plan:

• Design the interface.

• Configure the AI service.

• Implement input and response handling.

• Test different educational questions.

• Capture output and prepare project evidence.

Possible Challenges: Internet/service availability, response quality, prompt design, and secure handling of API credentials.

Evidence Placeholder: Insert the team's project planning/timeline screenshot here.

PHASE 8 – IDEATION OUTCOME & CONCLUSION

Final Ideation Outcome: The brainstorming process resulted in the EduGenie project concept, an AI-powered learning assistant designed to support interactive educational question answering.

Key Outcomes:

• Educational problem identified.

• Multiple solution ideas generated.

• Ideas discussed and refined.

• Final project concept selected.

• Target users and use cases identified.

• Initial feasibility and development plan prepared.

Future Scope: The idea can later be expanded with quizzes, subject-wise modes, multilingual support, voice interaction, progress tracking, and personalized learning features.

Conclusion: The Brainstorming & Ideation Phase established the problem, explored possible solutions, and provided a clear foundation for developing EduGenie.

Final Evidence Placeholder: Insert the final ideation summary/team discussion screenshot here.